

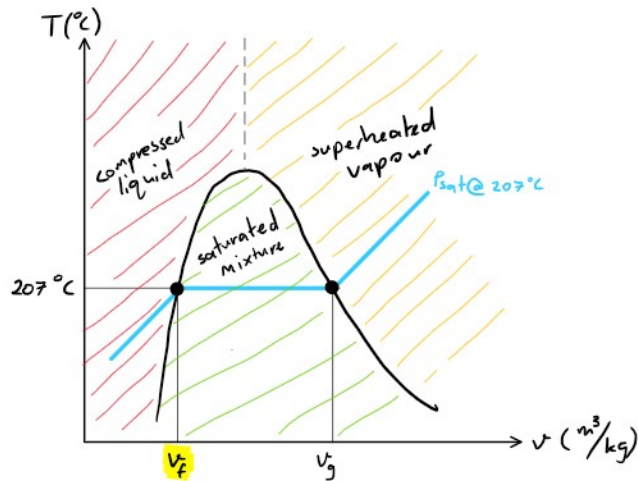
Interpolation (background)

Thursday, 7 August 2025 7:39 am

Example: Find the specific volume for saturated liquid water at 207°C

Givens

- 1) Phase: saturated liquid
 - 2) Temperature: 207°C
 - 3) We need to find v_f ← specific volume of a saturated liquid
- use table A-4 (saturated water table - temperature)



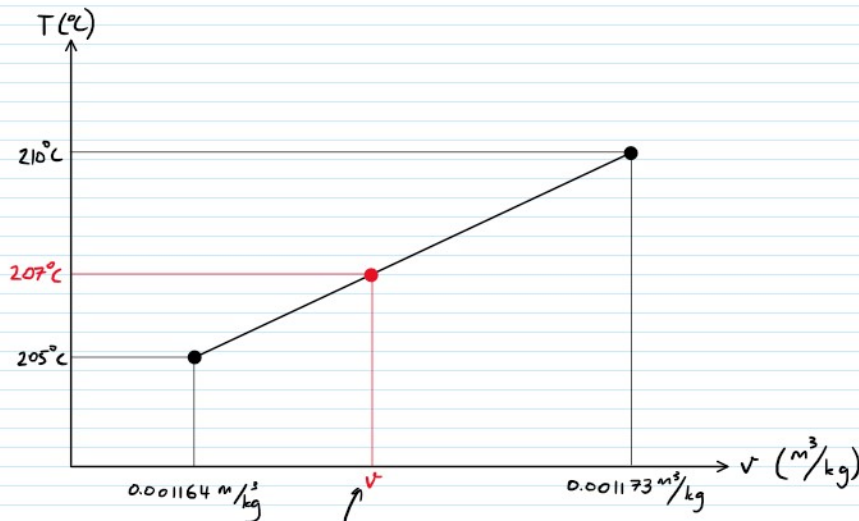
From table A-4:

Temp., T °C	Sat. press., P_{sat} kPa	Specific volume, m^3/kg	
		Sat. liquid, v_f	Sat. vapor, v_g
205	1724.3	0.001164	0.11508
210	1907.7	0.001173	0.10429

207°C

v

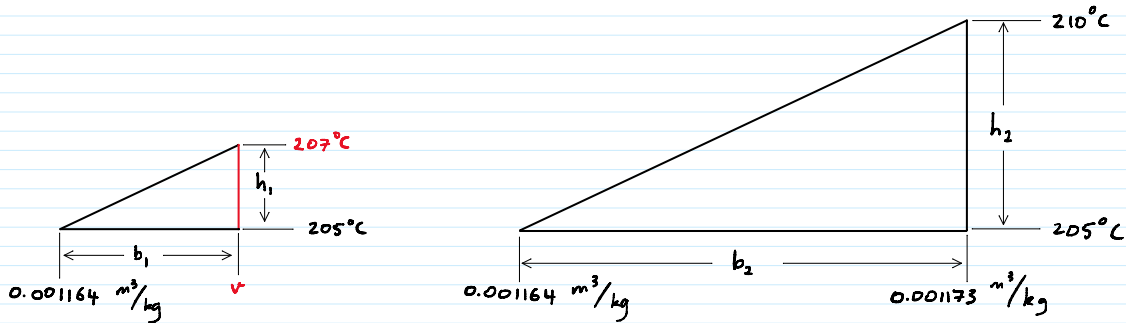
the value for v_f isn't given directly, so we'll need to approximate it using nearby data points



value that we're trying to find ($v_f@207°C$)

Using principle of similar triangles

Using principle of Similar triangles



height of small triangle $\frac{h_1}{b_1}$ = $\frac{h_2}{b_2}$ = height of big triangle
 base of small triangle b_1 base of big triangle b_2

the triangles are essentially the same shape, and are just scaled versions of each other, and so the ratios between their heights and bases are the same

$$\frac{207 - 205}{v - 0.001164} = \frac{210 - 205}{0.001173 - 0.001164} \quad \leftarrow \text{expanded equation}$$

$$\frac{2}{v - 0.001164} = \frac{5}{0.000009} \quad \leftarrow \text{simplified equation}$$

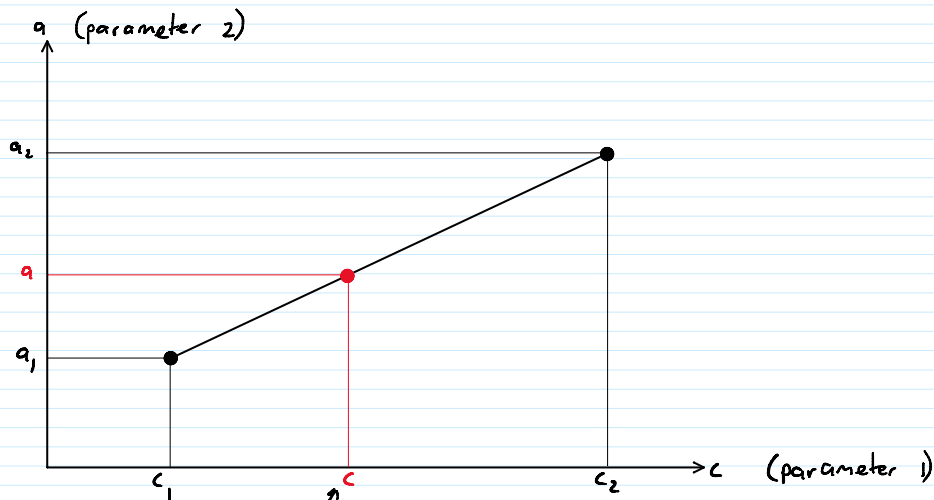
$$\frac{v - 0.001164}{2} = \frac{0.000009}{5}$$

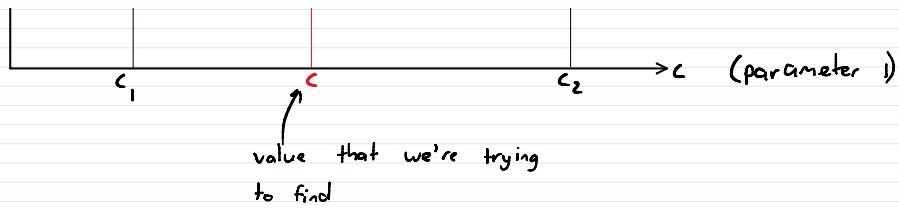
$$v - 0.001164 = 2 \left(\frac{0.000009}{5} \right)$$

$$v = 2 \left(\frac{0.000009}{5} \right) + 0.001164$$

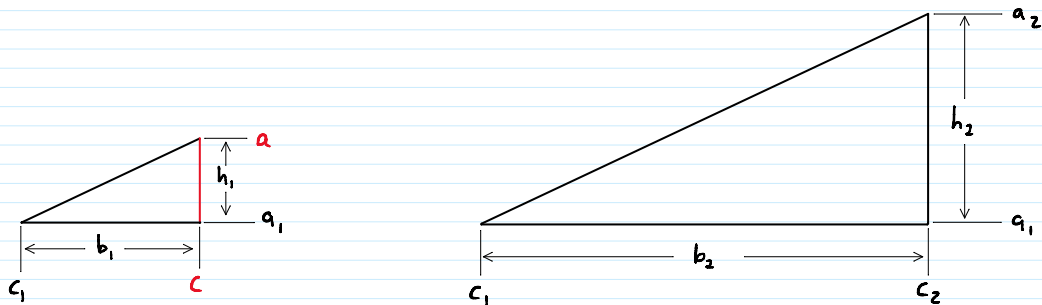
$$v = 0.0011676 \text{ m}^3/\text{kg} \quad \leftarrow \text{specific volume of saturated liquid water at } 207^\circ\text{C}$$

Deriving a general formula for interpolation





Using principle of similar triangles



$$\frac{h_1}{b_1} = \frac{h_2}{b_2}$$

$$\frac{a - a_1}{c - c_1} = \frac{a_2 - a_1}{c_2 - c_1}$$

$$\frac{c - c_1}{a - a_1} = \frac{c_2 - c_1}{a_2 - a_1}$$

$$c - c_1 = (a - a_1) \left(\frac{c_2 - c_1}{a_2 - a_1} \right)$$

$$c = c_1 + (a - a_1) \left(\frac{c_2 - c_1}{a_2 - a_1} \right) \leftarrow \text{general interpolation formula}$$

value you're trying to find (i.e. find c at the given value a)

given value

a_1 : value just below given value for a in the table
 a_2 : value just above given value for a in the table
 c_1 : value of c corresponding with a_1 in the table
 c_2 : value of c corresponding with a_2 in the table

Further practise

1. Find the specific internal energy of saturated liquid water (u_f) at 68°C

Temp., $T^\circ\text{C}$	Sat. press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg}\cdot\text{K}$		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898
60	19.947	0.001017	7.6670	251.16	2204.7	2455.9	251.18	2357.7	2608.8	0.8313	7.0769	7.9082
65	25.043	0.001020	6.1935	272.09	2190.3	2462.4	272.12	2345.4	2617.5	0.8937	6.9360	7.8296
70	31.202	0.001023	5.0396	293.04	2175.8	2468.9	293.07	2333.0	2626.1	0.9551	6.7989	7.7540

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2. Find the pressure of saturated water vapour that has a specific volume of $0.9 \text{ m}^3/\text{kg}$

Saturated water—Pressure table

Press., <i>P</i> kPa	Sat. temp., <i>T</i> _{sat} °C	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
		Sat. liquid, <i>v</i> _f	Sat. vapor, <i>v</i> _g	Sat. liquid, <i>u</i> _f	Evap., <i>u</i> _{fg}	Sat. vapor, <i>u</i> _g	Sat. liquid, <i>h</i> _f	Evap., <i>h</i> _{fg}	Sat. vapor, <i>h</i> _g	Sat. liquid, <i>s</i> _f	Evap., <i>s</i> _{fg}	Sat. vapor, <i>s</i> _g
175	116.04	0.001057	1.0037	486.82	2037.7	2524.5	487.01	2213.1	2700.2	1.4850	5.6865	7.1716
200	120.21	0.001061	0.88578	504.50	2024.6	2529.1	504.71	2201.6	2706.3	1.5302	5.5968	7.1270
225	123.97	0.001064	0.79329	520.47	2012.7	2533.2	520.71	2191.0	2711.7	1.5706	5.5171	7.0877
250	127.41	0.001067	0.71873	535.08	2001.8	2536.8	535.35	2181.2	2716.5	1.6072	5.4453	7.0525
275	130.58	0.001070	0.65732	548.57	1991.6	2540.1	548.86	2172.0	2720.9	1.6408	5.3800	7.0207

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